

Varunrao Wallajabad

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Education

University of Illinois Urbana-Champaign

Bachelor of Science in Computer Science

Champaign, IL

Expected May 2028

Technical Skills

Languages: C++, Python, Java, Bash, SQL, JavaScript, Rust

Frameworks & Libraries: Spring Boot, React, TensorFlow, Unity, Keras

Security Tools: Nmap, Burp Suite, Wireshark, Metasploit, GDB

Platforms & Tools: Git, Docker, Linux, Ubuntu, GCP, Firebase, GitHub Actions

Concepts: Memory Safety, Vulnerability Analysis, Secure Systems, Program Analysis, Object-Oriented Design

Experience

Cybersecurity Research Assistant

Jan 2026 – Present

University of Illinois Urbana-Champaign

Champaign, IL

- Audited 500+ security-critical C vulnerabilities within SARD and Defects4c datasets, documenting memory safety violations (CWE-119, CWE-416) to facilitate automated C-to-Rust transpilation research.
- Authored and validated exploit Proof-of-Concepts (PoCs) to verify vulnerability triggers and assess the efficacy of proposed memory-safe mitigation strategies.
- Benchmarked static and dynamic analysis tools (LLVM-based sanitizers) to optimize the detection of heap overflows and use-after-free errors in legacy codebases.

Assistant Tutor (CS 124: Intro to Computer Science)

Jan 2026 – Present

University of Illinois Urbana-Champaign

Champaign, IL

- Provide technical guidance to students on fundamental programming concepts in Java and Kotlin, focusing on object-oriented design and algorithmic problem-solving.
- Facilitate learning for 100+ students weekly through office hours and community forums, helping debug complex logic errors and explaining core computer science principles.
- Collaborate with course staff to identify common student misconceptions and improve instructional support materials.

Center Assistant

Aug 2022 – Present

Kumon North America

O'Fallon, MO

- Mentored 30+ students in advanced mathematics and literacy, translating complex academic concepts into accessible lessons tailored to diverse learning styles.
- Facilitated parent-teacher consultations to discuss student progress, demonstrating strong interpersonal communication and conflict-resolution skills.
- Collaborated with center leadership to streamline daily operations and student scheduling, ensuring a productive learning environment during peak hours.

Junior Software Developer

Oct 2021 – July 2022

UCode Inc.

Remote

- Architected interactive VR learning modules using Unity and C#, providing an immersive environment for K-12 students to visualize abstract computer science concepts.
- Optimized asynchronous rendering pipelines and mesh complexity, resulting in a 20% increase in frame rate stability on mobile VR hardware.

Awards & Honors

MORENET State Cybersecurity Competition — 3rd Place

2025

- Competed in live Red-Team/Blue-Team simulation, defending network infrastructure against 25+ teams.

Bay Area Hacks 2.0 — 1st Place

2024

- Developed a scalable software solution recognized for technical complexity among 500+ participants.

FBLA Website Coding & Development (State) — 2nd Place

2024

- Critiqued on technical stack choices, responsive design, and WCAG accessibility standards.

Projects

FinSight Analytics Platform | Java, Spring Boot, PostgreSQL, React, GCP

- Developed full-stack financial engine simulating equity clearing for 500+ trades across isolated containers.
- Integrated RESTful APIs with PostgreSQL to manage high-concurrency transactions with ACID compliance.
- Designed a dynamic React dashboard featuring real-time data visualization for portfolio risk-exposure.

UCode VR Education App | Unity, C#, Blender

- Conceptualized gamified logic puzzles in 3D to teach recursive logic and memory management fundamentals.

Campus Involvement

SigPwny, Association for Computing Machinery (ACM), Investment Portfolio Organization, SigEcom